

Volume 42

Sentinel Mobile Security Engine (SMSE)

Status: Core Security Domain

Priority: Critical

Vision

The Mobile Security Engine protects smartphones and tablets using the same architecture as every other Project Sentinel module.

The objective is not merely to identify risks.

The objective is to continuously understand the mobile device as part of the user's Digital Twin.

Supported Platforms

Initially:

- Android
- iOS
- iPadOS

Future architecture allows expansion to additional mobile operating systems if they emerge.

Mobile Philosophy

A phone is not just another endpoint.

It is often the user's:

- Identity device
- MFA device
- Banking device
- Crypto wallet
- Email client
- Camera
- Location device
- Authentication token
- Personal archive

Protecting the phone often protects the entire digital life of the user.

Mobile Object

Every phone becomes a Sentinel Object.

Organization

Mobile Identity

Every mobile device receives:

PS-MOBILE-00018492

Mobile Health Model

Rather than a single score.

Health consists of multiple dimensions.

Examples

Operating System

Security Updates

Application Risk

Permissions

Identity Protection

Encryption

Backup

Screen Lock

Network Security

Privacy

Trust

Evidence Freshness

Mobile Evidence

Examples

OS Version

Security Patch Level

Encryption Status

Screen Lock

Biometric Status

Developer Mode Status

Installed Applications (where permitted)

Application Permissions

Certificate Inventory

VPN Configuration

Wi-Fi Security

Bluetooth Status

Backup Configuration

Browser Configuration

Enterprise Management Status

Every observation becomes versioned evidence.

Mobile Application Engine

Applications become objects.

Each application records:

Developer

Publisher

Version

Permissions

Installation Date

Update History

Risk Classification

Evidence

Trust Contribution

Relationships

Permission Intelligence

One of our strongest consumer features.

Instead of displaying raw permissions.

Project Sentinel explains them.

Example

Camera

The emphasis is on transparency and informed decisions.

Mobile Configuration Drift

The platform continuously detects changes.

Examples

- New application installed
- Permissions modified
- VPN disabled
- Encryption disabled
- Screen lock removed
- Certificate added
- Enterprise profile changed

Everything becomes evidence.

Mobile Timeline

Example

Phone Purchased

Mobile Relationships

Phones rarely exist alone.

Example

Employee

This relationship model becomes extremely valuable during investigations.

Mobile Trust

Trust is calculated from evidence.

Examples

Positive indicators:

- Recent security patches
- Encryption enabled
- Strong authentication
- Verified backup
- Healthy configuration
- Stable security posture

Trust calculations remain explainable.

Personal Mobile Dashboard

Exactly as you envisioned.

The user opens Project Sentinel.

Immediately sees:

My Phone

Health

97%

Trust

98%

Protection

Enabled

Backup

Healthy

Applications

Reviewed

Identity

Protected

One screen.

Very easy to understand.

Enterprise Mobile Dashboard

Administrators can see:

Organization

↓

Country

↓

Department

↓

Team

↓

Phones

↓

Individual Device

Fleet management remains intuitive.

NEW CORE ENGINE

Mobile Privacy Engine

One of our signature modules.

It evaluates privacy-related settings that are observable through supported operating system capabilities.

Examples include:

- Excessive permission grants.
- Location access patterns.
- Background activity settings.
- Microphone and camera access.
- Clipboard access (where supported).

Recommendations are educational and evidence-based.

NEW CORE ENGINE

Mobile Identity Engine

Responsible for evaluating:

- Passkeys
- Biometrics

- MFA enrollment
- Authenticator configuration
- Device identity
- Enterprise identity integration

Because identity is now one of the most critical aspects of mobile security.

NEW CORE ENGINE

Mobile Recovery Engine

One of my favorite ideas.

It continuously evaluates recovery readiness.

Examples

Cloud Backup

Recovery Contact

Recovery Codes

Device Recovery

Account Recovery

Emergency Contacts

Home users rarely think about these until something goes wrong.

Project Sentinel helps them prepare in advance.

NEW CORE ENGINE

Mobile Companion Engine

A unique concept.

Instead of only protecting.

The application teaches.

Examples

"You haven't restarted your phone in several weeks. Installing pending system updates and rebooting can help ensure security fixes are applied."

"You have enabled automatic backups. Good job—this improves your ability to recover if your device is lost."

The platform encourages good security habits.

NEW IDEA

Hidden Application Awareness

This builds directly on one of your earliest requirements.

Some applications or components are intentionally difficult for users to notice.

Within the visibility and permissions provided by Android and iOS, Project Sentinel should help identify:

- Disabled-but-installed applications.
- Enterprise management profiles.
- Accessibility services with elevated privileges.
- Device administrator roles (where supported).
- Applications that no longer appear on the home screen but remain installed.
- Recently added profiles or certificates.

The engine reports observations transparently and explains what they mean, while respecting the security boundaries of the operating system.

NEW IDEA

Family Mobile Center

Parents can manage family devices through role-based permissions and explicit consent.

Examples

Parent

↓

Children

↓

Phones

↓

Tablets

↓

School Devices

Parents receive simple security summaries while respecting privacy boundaries appropriate to each role.

NEW IDEA

Mobile Security Passport

Every device receives a continuously updated passport.

Example

Phone

PS-ADR-0037

Mobile-First Security

Status: Accepted

Decision

Project Sentinel shall treat mobile devices as first-class security objects with dedicated evidence collection, relationship modeling, trust evaluation, and user guidance rather than as extensions of endpoint management.

Rationale

Modern digital identity, banking, communication, authentication, and personal data increasingly depend on mobile devices. A dedicated architecture better reflects their importance while remaining consistent with the Object Kernel and Digital Twin.

NEW LONG-TERM VISION

Personal Digital Life

I think we can now formally define another major vision for Project Sentinel.

For private users, the platform should protect **their entire digital life**, not just their devices.

That includes:

- Phone
- Laptop
- Tablet
- Home Wi-Fi
- Email accounts
- Cloud storage
- Password manager
- Passkeys
- Banking security
- Crypto wallets (configuration and device security, not custody)
- Smart home devices
- Backups
- Digital identity

The phone becomes the center of this personal Digital Twin.

One feature I believe could become iconic

Sentinel Security Timeline

Imagine opening the app and seeing:

Today

Instead of presenting cybersecurity as a collection of warnings, Project Sentinel tells the ongoing story of how the user's digital security is improving over time.

Looking Ahead

The next volume will be **Volume 43 - Sentinel Identity Security Engine (SISE)**.

In my opinion, identity will eventually become **the single most important security domain**. Whether the protected asset is a bank, a government agency, a crypto exchange, or a home user, nearly every attack ultimately targets identities, credentials, sessions, or authorization. Designing this engine correctly will make the rest of Project Sentinel significantly stronger because every other module—from Endpoint and Mobile to Cloud and Web—depends on trusted identity. I believe it will become one of the cornerstone capabilities of the entire platform.

=====[USER]=====

Good and continue

=====[ASSISTANT]=====

Excellent. I think we are now designing one of the most important security engines in the entire platform.

If I had to choose only one engine to protect an organization today, it would probably be **Identity Security**.

Devices can be replaced.

Servers can be rebuilt.

Applications can be redeployed.

But a compromised identity can bypass many traditional security controls.

That is why Identity must become a first-class citizen in Project Sentinel.

PROJECT SENTINEL BLUEPRINT

Volume 43